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One-Bit Quantization with Sub-Nyquist Sampling: A Low-Complexity Receiver Design in ISAC System

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***Abstract*—In this paper, a low-complexity receiver is designed for low-priced hardware implementation in the integrated sensing and communication (ISAC) system. The receiving antennas are sparsely located, and the signal is sampled with low-resolution and low-sampling rate analog-to-digital converters (ADCs). An orthogonal frequency division multiplexing-based ISAC waveform is considered where preambles are used for target sensing. Range, Doppler and direction-of-arrival are estimated via a block-sparse quantized orthogonal matching pursuit (BS-QOMP) method. Numerical results show that the estimation performance of the proposed scheme degrades 8.5 dB compared with the conventional design, while having a 74 dB reduction in power consumption. And the localization error under the practical base station (BS) setting is 0.2 m for the target located 400 m away from the BS.**

***Index Terms*—Integrated sensing and communication, one-bit quantization, sub-Nyquist sampling, quantized orthogonal matching pursuit, low-complexity receiver.**

I. Introduction

Integrated sensing and communication (ISAC) is a promising technique to solve the problem of spectrum congestion, aiming at sharing spectrum between wireless communication and radar sensing [1]. Following the development of this technology, applications with sensing capabilities have attracted widespread attentions like smart home, environment monitoring and autonomous driving [2], [3]. One key scenario is the perceptive mobile network, where base stations (BSs) are required to implement both radar and communication functionalities on the same platform.

However, when the target is near the BS, the reflected waveform emitted by the panel is unable to be received during transmitting, resulting in a blind area around the BS. To address this problem, an additional receiving panel used for sensing is required to achieve a full-duplex mode. In addition, a multiple input multiple output orthogonal frequency division multiplexing (MIMO-OFDM) system is considered to achieve high-resolution sensing. Although the large-scale antenna system and the large bandwidth OFDM signal guarantee high-resolution in angle and range estimates, a great number of analog-to-digital converters (ADCs) with high sampling rates are required, which makes the ADC energy-intensive and expensive [4], [5]. In this case, it is vital to design an energy-saving and cost-efficient architecture which can be widely implemented in the mobile network [6].

The cost of ADC can be lowered by decreasing the sampling rate [7], [8] and quantizing with less bits [9], [10]. In [11], a sub-Nyquist sampling based OFDM radar is considered. The radar with reduced hardware complexity and lower power consumption is achieved while the noise level is increased due to the spectrum aliasing. On the other hand, one-bit quantization, as an extreme form of quantization, can be simply accomplished by comparing the signal with a reference level and representing the signal with only one bit [12]. In this case, the procedure of sampling and quantizing becomes easily accomplished, making the system cheap and affordable [13]. In addition, by applying the sparse antenna array, less antennas are utilized with same angle resolution, and the number of ADC is reduced at the same time.

Although the hardware is simplified with these designs, a more generic model that can be designed flexibly is needed. Furthermore, an efficient and low-complexity algorithm under this framework is required for target sensing in multiple dimensions. And there is a lack of performance evaluation under the actual BS setting. Motivated by this, in this paper, we propose a new receiver design combining one-bit quantization and sub-Nyquist sampling together to lower the price and complexity of the receiver. The sparse array is also considered to further simplify the architecture. A sparse estimation algorithm called block-sparse quantized orthogonal matching pursuit (BS-QOMP) is proposed to solve the block-sparse signal recovery problem. Simulations show that around 74 dB reduction in energy consumption is achieved at a cost of 8.5 dB in performance loss. Besides, under the actual BS setting, the localization error is 0.2 m for the target 400 m away from the BS.

The rest of the paper is organized as follows: Section II introduces the system model and block diagram. Section III explains the one-bit quantization and sub-Nyquist sampling. Section IV presents the detailed algorithm. Simulations and analysis are given in Section V. Finally, Section VI concludes the whole paper.

II. System Model

Consider an active sensing scenario, the BS sweeps the transmitter beam for target sensing, as shown in Fig. 1. An additional panel is installed below the transmitter and receives the reflected ISAC signal with a sparse array, which is illustrated in Fig. 2. The receiving antennas on the panel are randomly located from 0 to D with the uniform distribution, where D is the size of the array. For a general uniform linear array, the space of two adjacent elements should be equal or less than half of the wavelength λ in avoid of angle ambiguity. In this case, D is equal to $M_r\lambda/2$ where M_r is the number of the receiving antenna. However, the size is enlarged by ratio β_a for sparse array, which is represented as $\beta_a M_r \lambda/2$. The locations of elements are donated as $d_0, \ldots, d_{M_r-1}$ and the direction-of-arrival (DOA) for the k-th target is θ_k. In this case, the steering vector of the sparse array is written as $\mathbf{b}(\theta_k) = [e^{-j2\pi d_0 \sin(\theta_k)/\lambda}, \ldots, e^{-j2\pi d_{M_r-1} \sin(\theta_k)/\lambda}]^T \in \mathbb{C}^{M_r \times 1}$, where zero is chosen as the reference point.

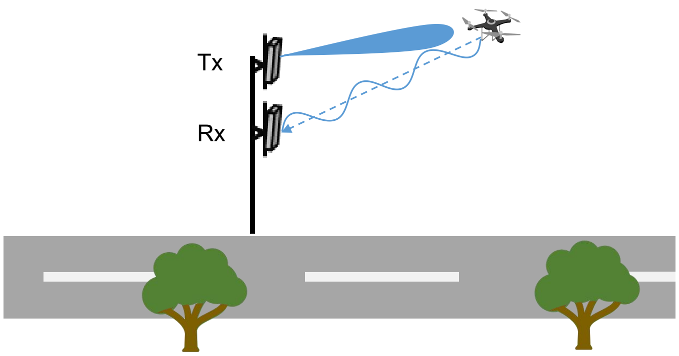

Fig. 1. System model.

After being received by the array, the signals are amplified by M_r low-noise-amplifiers (LNAs) and down-converted to the baseband with a local oscillator. Then, they are uniformly sampled below the Nyquist rate and quantized into one-bit digital signals by comparing with the zero threshold at ADCs. Finally, the one-bit measurements are processed at the digital signal processing block to estimate the unknown parameters.

Assume the transmitting signal is OFDM-based and the pilots in preambles are complex-Gaussian distributed, i.e. $x_{n,m} \sim \mathcal{CN}(0, P_s)$, where $x_{n,m}$ is the symbol on the n-th subcarrier of the m-th preamble signal, P_s is the transmit power. Preambles are used for sensing and the interval between two preambles is defined as T_p. Therefore, the m-th preamble signal is expressed as

$$s(t,m) = \sum_{n=0}^{N_c-1} x_{n,m} e^{j2\pi(f_c+n\Delta f)t} \mathrm{Rect}\left(\frac{t-mT_p}{T_s+T_c}\right), \quad (1)$$

where N_c is the number of subcarriers, f_c is the carrier frequency, Δf is the subcarrier interval, $T_s = 1/\Delta f$ is duration of the OFDM symbol, and $\mathrm{Rect}(\frac{t}{T_s+T_c})$ is the rectangle window function of length $T_s + T_c$, corresponding to the period of OFDM symbol with cyclic prefix (CP).

Assume there are K targets in the sweeping area, then the

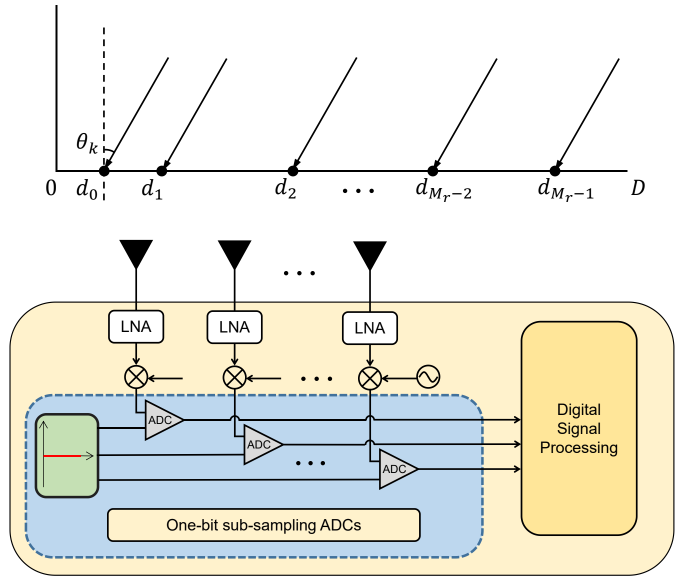

Fig. 2. Sparse array and block diagram.

channel response can be represented as

$$\mathbf{h}(t) = \sum_{k=0}^{K-1} \tilde{\alpha}_k \delta(t-\tau_k) G_t \mathbf{b}(\theta_k), \quad (2)$$

where $\tilde{\alpha}_k$ and τ_k are the complex channel gain and signal propagation delay of the k-th target, and G_t is the antenna gain from the transmitter. By doing convolution with equation (1), the noise-free vector of received signal can be written as

$$\begin{aligned} \tilde{\mathbf{y}}(t,m) &= s(t,m) \otimes \mathbf{h}(t) \\ &= \sum_{k=0}^{K-1} \tilde{\alpha}_k s(t-\tau_k, m) G_t \mathbf{b}(\theta_k) \\ &= \sum_{k=0}^{K-1} \sum_{n=0}^{N_c-1} \tilde{\alpha}_k x_{n,m} e^{j2\pi(f_c+n\Delta f)(t-\tau_k)} G_t \mathbf{b}(\theta_k), \end{aligned} \quad (3)$$

where rectangle window function Rect is omitted for simplicity, as we assume the multi-path delay is in the duration of CP. Furthermore, τ_k can be rewritten as $2(r_k - v_k t)/c$, where r_k is the range between the BS and the k-th target, v_k is the radical velocity of the k-th target and c is the speed of light. Because the coherent processing interval is small, we assume r_k is fixed during the whole period. Based on that, equation (3) can be further written as

$$\begin{aligned} &\tilde{\mathbf{y}}(t,m) \\ &= \sum_{k=0}^{K-1} \sum_{n=0}^{N_c-1} \alpha_k x_{n,m} e^{j2\pi(f_c+f_{D,k})t} e^{j2\pi n\Delta f(t-\tau_k)} \mathbf{b}(\theta_k), \end{aligned} \quad (4)$$

where $\alpha_k = \tilde{\alpha}_k e^{-j4\pi f_c r_k/c} G_t$, and $f_{D,k} = 2v_k f_c/c$ is the Doppler shift of the k-th target. After down-conversion, the baseband signal with noise is expressed as

$$\begin{aligned}&\mathbf{y}(t,m)\\&=\sum_{k=0}^{K-1}\sum_{n=0}^{N_c-1}\alpha_k x_{n,m}e^{j2\pi f_{D,k}t}e^{j2\pi n\Delta f(t-\tau_k)}\mathbf{b}(\theta_k)+\mathbf{w}(t),\end{aligned} \tag{5}$$

where $\mathbf{w}(t) \in \mathbb{C}^{M_r\times 1}$ is the complex additive-white-Gaussian-noise vector. The received signals shown in (5) are converted to digital signals by ADCs for further digital processing. In order to ease the burden of ADC, sub-Nyquist sampling and one-bit quantization are applied in the following sampling and quantization procedure.

III. Sub-sampling and One-Bit Quantization

In this section, we introduce sub-sampling and one-bit quantization, the sparse representation of the quantized signal is also derived for further processing.

According to the sampling theory, it is well known that for a bandlimited OFDM signal, in order to recover the signal undistorted, the minimum sampling frequency is $B = N_c\Delta f$, termed the Nyquist rate [7]. However, sub-sampling gives a way to sample the signal under the Nyquist rate, trying to use fewer samples to recover the signal, which is widely discussed in compressive sensing (CS) [14], [15]. In this paper, the continuous signals received by antennas are sub-sampled by uniformly sampling ADCs with the sub-Nyquist sampling rate B/β_s, where β_s is the sub-sampling ratio [11]. Signals in equation (5) are sampled with the period of $\beta_s/\Delta f N_c$. Therefore, the discrete signal vector of the p-th sampling point in the m-th preamble can be written as

$$\begin{aligned}&\mathbf{y}_{p,m}\\&=\sum_{k=0}^{K-1}\alpha_k e^{j2\pi m T_p f_{D,k}}\mathbf{b}(\theta_k)\sum_{n=0}^{N_c-1}x_{n,m}e^{j2\pi n\Delta f(p\frac{\beta_s}{\Delta f N_c}-\tau_k)}\\&+\mathbf{w}_{p,m}, p\in\{0,\dots,P-1\}, m\in\{0,\dots,M_s-1\},\end{aligned} \tag{6}$$

where $\mathbf{w}_{p,m}$ is the sub-sampled noise vector from $\mathbf{w}(t)$ with the period of $\beta_s/\Delta f N_c$, following $\mathcal{CN}(\mathbf{0},\sigma^2\mathbf{I}_{M_r})$, and $P = \lfloor N_c/\beta_s\rfloor$ is the number of sampling points in an OFDM symbol. Although the sampling burden of ADCs is released, sub-Nyquist sampling also introduces the sub-band folding of the OFDM signal, making modulated pilots on different subcarriers overlap in the frequency domain [11]. For example, when $\beta_s = 2$, the pilot $x_{m,n}$ in frequency domain is transformed to time domain signal with $N_c/2$-points inverse discrete Fourier transform, shown as

$$\begin{aligned}\mathcal{F}^{-1}\{x_{p,m}\}&=\sum_{n=0}^{N_c-1}x_{n,m}e^{j2\pi p\frac{2n}{N_c}}\\&=\sum_{n=0}^{\frac{N_c}{2}-1}x_{n,m}e^{j2\pi p\frac{2n}{N_c}}+\sum_{n=\frac{N_c}{2}}^{N_c-1}x_{n,m}e^{j2\pi p\frac{2n}{N_c}},\end{aligned} \tag{7}$$

resulting in the overlapping of $x_{0:N_c/2-1,m}$ and $x_{N_c/2:N_c-1,m}$ in frequency domain.

By stacking all samples from M_s OFDM symbols and M_r antennas into a matrix $\mathbf{Y} \in \mathbb{C}^{PM_s\times M_r}$, the signal can be rewritten as

$$\mathbf{Y}=\sum_{k=0}^{K-1}\alpha_k\mathbf{a}(\tau_k,f_{D,k})\mathbf{b}^T(\theta_k)+\mathbf{W}, \tag{8}$$

where $\mathbf{W} = [\mathbf{w}_{0,0},\mathbf{w}_{1,0},\dots,\mathbf{w}_{P-1,M_s-1}]^T \in \mathbb{C}^{PM_s\times M_r}$ is the noise matrix. The time domain vector $\mathbf{a}(\tau_k, f_{D,k})$ in equation (8) is donated as

$$\mathbf{a}(\tau_k,f_{D,k})=\mathrm{vec}\left(\mathbf{v}^T(f_{D,k})\odot\mathbf{CR}(\tau_k)\right), \tag{9}$$

where $\mathbf{v}(f_{D,k}) = [1, e^{j2\pi T_p f_{D,k}},\dots,e^{j2\pi(M_s-1)T_p f_{D,k}}]^T$ is the Doppler steering vector, and $\mathbf{C} \in \mathbb{C}^{P\times N_c}$ is the sampling matrix, which is given as

$$\mathbf{C}=\begin{bmatrix}1 & 1 & \cdots & 1\\ 1 & e^{j2\pi\beta_s/N_c} & \cdots & e^{j2\pi\beta_s(N_c-1)/N_c}\\ \vdots & \vdots & \ddots & \vdots\\ 1 & e^{j2\pi\beta_s(P-1)/N_c} & \cdots & e^{j2\pi\beta_s(P-1)(N_c-1)/N_c}\end{bmatrix}, \tag{10}$$

$\mathbf{R}(\tau_k) = \mathbf{r}(\tau_k)\odot\mathbf{X} \in \mathbb{C}^{N_c\times M_s}$ is the delay encoded matrix, where $\mathbf{r}(\tau_k) = [1, e^{-j2\pi\Delta f\tau_k},\dots,e^{-j2\pi(N_c-1)\Delta f\tau_k}]^T$ is the delay steering vector and $\mathbf{X}$ is the pilot matrix with $x_{n,m}$ in the n-th row and m-th column.

This equation can be further represented in sparse formation

$$\mathbf{Y}=\mathbf{AB}+\mathbf{W}, \tag{11}$$

where $\mathbf{A} \in \mathbb{C}^{PM_s\times M_\tau M_f}$ is the dictionary matrix comprising columns of $\mathbf{a}(\tau_i^g, f_{D,j}^g)$ with different grids for estimation, i.e.

$$\mathbf{A}=[\mathbf{a}(\tau_0^g,f_{D,0}^g),\mathbf{a}(\tau_1^g,f_{D,0}^g),\dots,\mathbf{a}(\tau_{M_\tau-1}^g,f_{D,M_f-1}^g)], \tag{12}$$

where M_τ and M_f are the numbers of grid points for delay τ and Doppler shift f_D. Assume the grid is fine enough so that the underestimated parameters are located on it. We define $L = M_\tau M_f$ as the total grid number, and the l-th column of $\mathbf{A}$ is represented as $\mathbf{a}_l$. The estimation grids for DOA are also generated and donated by θ_q^g with M_θ grid points, thus the steering vector of the grid is donated as $\mathbf{b}(\theta_q^g)$. Assume all these grids are uniformly spaced.

Notice the parameter θ is not considered in the dictionary matrix $\mathbf{A}$, as too many underestimate parameters would enlarge the size of $\mathbf{A}$, which increases the data volume of the dictionary matrix and slows down the speed of the algorithm. In addition, the grid is able to be finer with less parameters, which is beneficial for sparse reconstruction. In this case, the sparse vector is transformed into a row-sparse matrix $\mathbf{B} \in \mathbb{C}^{M_\tau M_f\times M_r}$ with K non-zero rows, which can be represented explicitly as

$$\mathbf{B}_{i,j}^{\mathrm{row}}=\begin{cases}\alpha_k\mathbf{b}^T(\theta_k), & \tau_i^g=\tau_k, f_{D,j}^g=f_{D,k},\\ 0, & \text{otherwise},\end{cases} \tag{13}$$

where the DOA of targets are encoded into the phase of non-zero vectors, and $\tilde{\mathbf{b}}_l^T = \alpha_k\mathbf{b}^T(\theta_k)$ is the l-th row of $\mathbf{B}$.

After sub-sampling, the baseband signal $\mathbf{Y}$ is quantized into one-bit data via comparing with the zero threshold, which is detailed as

$$\overline{\mathbf{Y}} = \mathcal{Q}(\mathbf{Y}) = \text{sign}(\Re\{\mathbf{Y}\}) + j\text{sign}(\Im\{\mathbf{Y}\}), \tag{14}$$

where $\mathcal{Q}(\cdot)$ is the quantization function, $\text{sign}(\cdot)$ is the comparing function with $\text{sign}(x) = 1$ for $x > 0$ and $\text{sign}(x) = -1$ otherwise. A method is proposed to estimate the parameters based on the quantized measurements.

IV. Block-sparse Quantized Orthogonal Matching Pursuit

Methods for CS have been explored for decades. It is well known that the orthogonal matching pursuit (OMP) algorithm is widely used in sparse signal recovery for its low complexity, [16] extends the continuous OMP to quantized observations, proving its effectiveness for one-bit data. However, the method in [16] only considers the estimation of velocity for a simple continuous wave radar, and the quantized atom can not be removed thoroughly when updating the residue because of the amplitude information loss for one-bit quantization. In this section, a BS-QOMP algorithm is proposed to estimate the delay, Doppler and DOA of targets for OFDM signal. A row-sparse matrix is constructed to reduce the size of the dictionary for memory saving, and an amplitude coefficient is introduced to calculate the residue of the one-bit signal.

It is verified that the projected back-projection algorithm is efficient with the one-bit measurement [17], [18]. Thus, by calculating the contributions from different bases, the first non-zero row could be found, which is shown as

$$\hat{l} = \underset{l \in \{1,\dots,L\}}{\arg\max} \|\mathbf{a}_l^H \overline{\mathbf{Y}}\|_1, \tag{15}$$

where $\|\cdot\|_1$ is the $\mathcal{L}_1$-norm to enforce a row-sparse solution. After obtaining the maximum column $\hat{l}$, the coefficients of $\tilde{\mathbf{b}}_{\hat{l}}^T$ can be estimated approximately with the equation

$$\tilde{\mathbf{b}}_{\hat{l}}^T = \underset{\tilde{\mathbf{b}}_{\hat{l}}^T \in \mathbb{C}^{M_r}}{\arg\min} \|\mathbf{a}_{\hat{l}} \tilde{\mathbf{b}}_{\hat{l}}^T - \overline{\mathbf{Y}}\|_2^2, \tag{16}$$

the equation can be solved by calculating the pseudo-inverse of vector $\mathbf{a}_{\hat{l}}$, which is detailed as $\tilde{\mathbf{b}}_{\hat{l}}^T = \mathbf{a}_{\hat{l}}^\dagger \overline{\mathbf{Y}}$, where $\mathbf{a}_{\hat{l}}^\dagger = \mathbf{a}_{\hat{l}}^H / \|\mathbf{a}_{\hat{l}}\|_2^2$ is the left inverse of the vector. Then, the corresponding part after quantization is derived via

$$\tilde{\mathbf{Y}} = \mathcal{Q}(\mathbf{a}_{\hat{l}} \tilde{\mathbf{b}}_{\hat{l}}^T), \tag{17}$$

and the residue is computed as

$$\overline{\mathbf{Y}} = \overline{\mathbf{Y}} - \epsilon \tilde{\mathbf{Y}}, \tag{18}$$

where $\epsilon = \|(\mathbf{A}^H \overline{\mathbf{Y}})^T\|_1 / \|(\mathbf{A}^H \tilde{\mathbf{Y}})^T\|_1$ is the amplitude coefficient of the component for updating the residue, as the one-bit quantization omits the amplitude information and changes the energy of the omitted part. After K iterations, K targets are found and the corresponding parameters can be derived from the non-zero index of the sparse matrix. Furthermore, by matching the steering vector of the grid $\mathbf{b}(\theta_q^g)$ with $\tilde{\mathbf{b}}_{\hat{l}}^T$, the DOAs of targets are derived after searching the maximum peaks of the pattern. The detailed algorithm is presented in Algorithm 1.

Algorithm 1: BS-QOMP

Input: The one-bit data $\overline{\mathbf{Y}}$; The dictionary matrix $\mathbf{A}$; The target number K.
Output: Range d_k, radical speed v_k and DOA θ_k.

for $k = 1$ *to* K **do**
  Derive the desired index according to (15) by searching the maximum peak;
  Calculate the coefficients of the non-zero vector via (16);
  Remove the corresponding part and derive the residue with (17) and (18);
end
Find d_k and v_k from $\hat{l}$;
Match $\mathbf{b}(\theta_q^g)$ with $\tilde{\mathbf{b}}_{\hat{l}}^T$ and search the maximum peak index θ_k;

V. Simulation Results

In this section, we evaluate the performance of the BS-QOMP method and the effects of one-bit quantization and sub-Nyquist sampling. The ISAC waveform is OFDM-based and the carrier frequency is 2.6 GHz. The bandwidth of the OFDM signal is 100 MHz and the subcarrier interval is 30 kHz. There are 10 preambles used for sensing and the symbol period is 2.5 ms. Assume there are 3 targets in the model, and the underestimated parameters, i.e. range (m), radical velocity (m/s) and DOA (rad), are [54.5, 72.2, 101.8], [1.8, -7.4, 9.2] and [-1.0472, 0.5236, 0.7854], respectively. The signal-to-noise ratio (SNR) of the signal is defined as $SNR = P_s N_c / \sigma^2$, where P_s is the average power of the modulated symbol in one subcarrier, thus $P_s N_c$ is the transmit power of the multi-carrier signal, i.e. OFDM signal in time domain, and σ^2 is the variance of the noise at the receiver side. In addition, we assume the magnitudes of path loss α_k are [0.8660, 0.4330, 0.5774] with random phase uniformly from 0 to 2π. The total grid number $L = 12000$ with $M_\tau = 300$ for delay and $M_f = 40$ for Doppler shift.

A. *Performance Evaluation for Low-complexity Receiver*

Fig. 3 depicts the root-mean-square error (RMSE) of distance, speed and angle estimation, defined as

$$RMSE = \sqrt{\frac{1}{K} \sum\nolimits_{k=0}^{K-1} \|\Delta_k\|^2}, \tag{19}$$

where Δ_k is the estimation error of parameters, which is calculated with the closest result. Four different schemes are considered for comparison: **(a)** conventional: conventional model without one-bit quantization and sub-Nyquist sampling, $M_r = 20$ receiving antennas are considered with the interval of $\lambda/2$, angles are estimated first via multiple signal classification (MUSIC) algorithm and equal gain combining is applied to strength the signal, after that delay and Doppler are

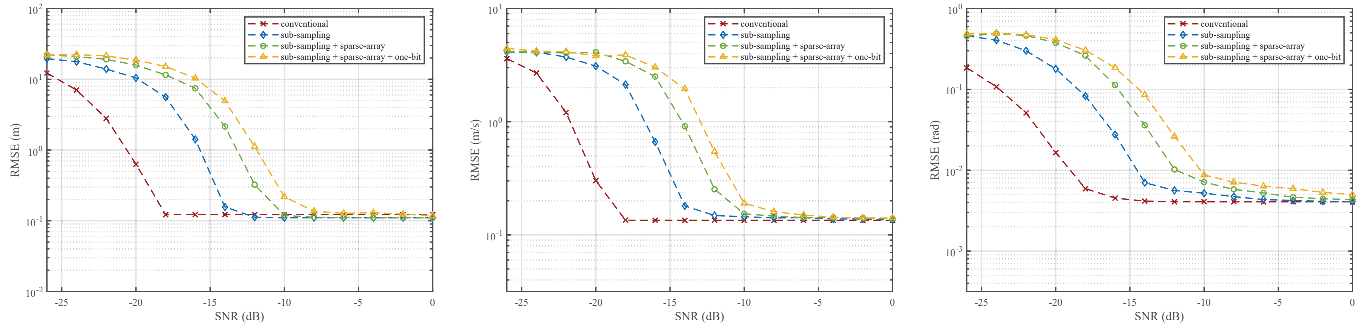

Fig. 3. Sensing performance vs. SNR. for distance (left), speed (middle) and angle (right) estimates

estimated with the fast Fourier transform (FFT) periodogram. **(b)** sub-sampling: sub-Nyquist sampling is considered with the rate $\beta_s = 100$ and the receiving antenna number is 20 like conventional model, the OMP method is used for estimation. **(c)** sub-sampling + sparse-array: same sub-sampling rate is considered, receiving antennas are randomly located and reduced to 10 with same array size. **(d)** sub-sampling + sparse-array + one-bit: based on scheme (c), one-bit quantization is further considered and the BS-QOMP algorithm is applied.

The figures demonstrate the performance loss from sub-sampling, sparse-antenna and one-bit quantization. Under the sub-sampling scheme, the noise is enlarged by 20 dB with the sub-sampling rate of 100, causing around 4 dB performance loss, while the sampling rate of ADC is reduced to 1% in return and less samples are processed in the following processing. By applying the sparse array and decreasing the antenna number from 20 to 10, the accuracy of angle estimation is maintained, while half of the ADCs are saved in price of 3 dB performance loss. Furthermore, there is only 1.5 dB loss with one-bit quantization, while ADC can be simply implemented like a comparator.

B. Performance Evaluation Under Practical BS Settings

Fig. 4 illustrates the performance of target localization as function of BS-target distance. The radar cross-section (RCS) of the target is equal to $\sigma_{RCS} = 0.01$ m^2, the effective radiated power of the BS is $P_t G_t = 44$ dBm. Besides, the noise power is $\sigma^2 = kTFB$, where $k = 1.38 \cdot 10^{-23}$ JK^{-1} is the Boltzmann's constant, $T = 290$ K is the reference temperature, and $F = 8$ dB is the receiver noise figure. A single target is considered in this scenery, it is shown that under the proposed framework, the positioning RMSE is 0.2 m when sensing a target 400 m away from the BS.

C. Power Consumption and Algorithm Complexity

The power consumption is also investigated. According to [5], the power dissipation of an ideal ADC is derived as

$$P_{\min} = k \cdot T \cdot f_s \cdot 10^{(6N+1.76)/10}\text{W}, \tag{20}$$

where f_s is the sampling rate and N is the bit-depth. Although it is stated that the relationships of sampling rate and bit number are not independent from each other, and the power

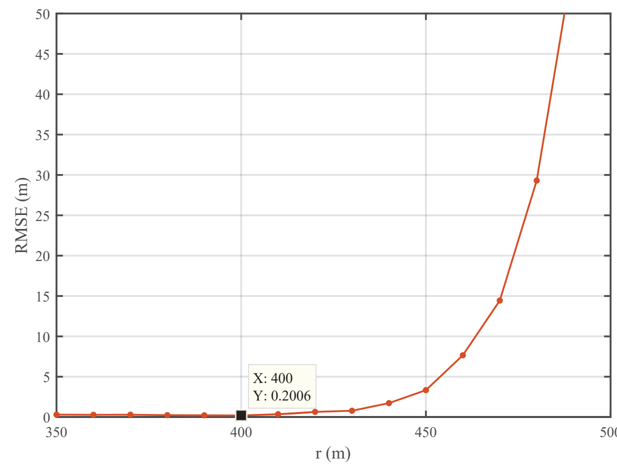

Fig. 4. Performance of target localization vs. BS-target distance.

dissipation is strongly related to the structure of ADC, we can still get a glimpse of the benefits of sub-Nyquist sampling and one-bit quantization: ADC is simplified and less power is consumed. In this paper, the sampling rates f_s for the conventional model and the proposed model are 100 MHz and 1 MHz, respectively. Thus, only 1% power is consumed with sub-Nyquist sampling according to equation (20). In addition, we assume the bit-depth N for the conventional model is 10 and the reduction of power is 54 dB in the one-bit framework. The overall power dissipation can be calculated by substituting k and T in equation (20), which is detailed in Table I. By comparing and calculating the ratio of the power consumption for scheme (a) and (d), a 74 dB power saving is obtained.

Finally, we evaluate the complexity of algorithms. The computation cost for BS-QOMP is proportional to $\mathcal{O}(KPM_sM_rL)$, where PM_sM_rL comes from the matrix multiplication $\mathbf{A}^H\overline{\mathbf{Y}}$, and K is the iteration number. In the first iteration, $\overline{\mathbf{Y}}$ is one-bit data, so the multiplication is degraded to the summation. The space complexity is proportional to $\mathcal{O}(PM_sL)$, as the dictionary matrix $\mathbf{A}$ needs to be stored. Besides, the cost for MUSIC is $\mathcal{O}(M_r^3)$ and that for FFT is $\mathcal{O}(M_sN_c\log(M_sN_c))$. Therefore, the figure for the conventional scheme is $\mathcal{O}(M_r^3 + M_sN_c\log(M_sN_c))$, the results are also summarized in Table I.

TABLE I
SCHEME ANALYSIS

Scheme	Number of ADC	Power (W)	Algorithm Complexity
Conventional	20	6.0017×10^{-7}	$\mathcal{O}(M_r^3 + M_s N_c \log(M_s N_c))$
Sub-sampling	20	6.0017×10^{-9}	$\mathcal{O}(KPM_s M_r L)$
Sub-sampling+sparse-array	10	6.0017×10^{-9}	
Sub-sampling+sparse-array+one-bit	10	2.3893×10^{-14}	

In order to further lower the algorithm complexity and the storage burden for hardware implementation, the elements in the dictionary matrix $\mathbf{A}$ can be further quantized into 2-bit data, i.e. one performs as the sign bit and another representes the value. The result is shown in Fig. 5 under the same simulation condition listed before. Distance estimation is taken as an example, it is shown that limited loss is seen, while substantial calculation spaces and resources are saved in this manner.

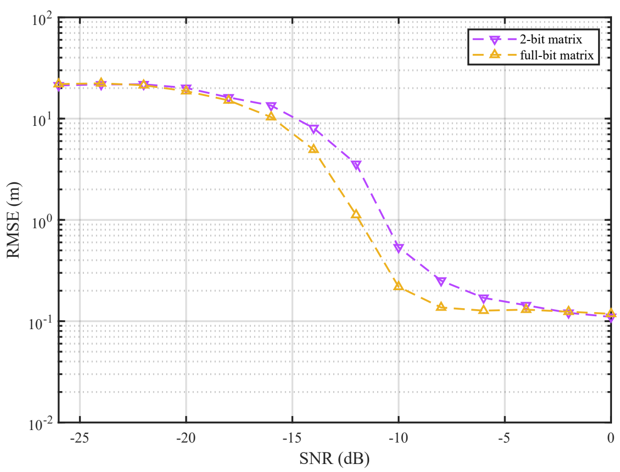

Fig. 5. Distance sensing performance under 2-bit dictionary matrix.

VI. CONCLUSIONS

In this paper, a low-complexity receiver design was proposed, one-bit quantization and sub-Nyquist sampling were considered for low-power consumption and price-friendly ADC. A one-bit CS problem is formulated and solved with the BS-QOMP algorithm. Simulations show its effectiveness and illustrate the gap between the proposed design and the traditional receiver. It is demonstrated that our scheme suffers from an 8.5 dB performance loss, but gains a 74 dB energy improvement. With an actual setting, the target on 400 m can be localized with the RMSE of 0.2 m using the proposed receiver.

VII. ACKNOWLEDGMENT

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